

Winning Space Race with Data Science

https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Collected Falcon 9 launch data via the SpaceX REST API (and IBM's static snapshot after persistent live-API outages) and by web scraping the Falcon 9/Heavy Wikipedia launch table; wrangled it into a clean, analysis-ready dataset with an engineered landing-success label
- Explored the data with visualizations (payload, orbit, launch site, and yearly trends), SQL queries, an interactive Folium map and a Plotly Dash dashboard; then trained and tuned four classification models (Logistic Regression, SVM, Decision Tree, KNN) via GridSearchCV
- The Decision Tree classifier performed best, with a cross-validated accuracy of 86.25%, showing that first-stage landing outcomes can be predicted with good reliability from launch parameters such as payload mass, orbit, and launch site

Introduction

- SpaceX advertises Falcon 9 rocket launches at a cost of $62 million, considerably lower than competitors, largely because SpaceX can reuse the first stage rather than discarding it. If we can determine whether the first stage will land successfully, we can estimate the cost of a launch – information valuable to a company bidding against SpaceX
- This project asks: can we predict, from publicly available launch data (payload mass, orbit type, launch site, booster version, and flight history), whether the Falcon 9 first stage will land successfully? Which factors most influence a successful landing, and which combination of these factors produces the highest success rate?

Section 1

Methodology



Methodology

Executive Summary

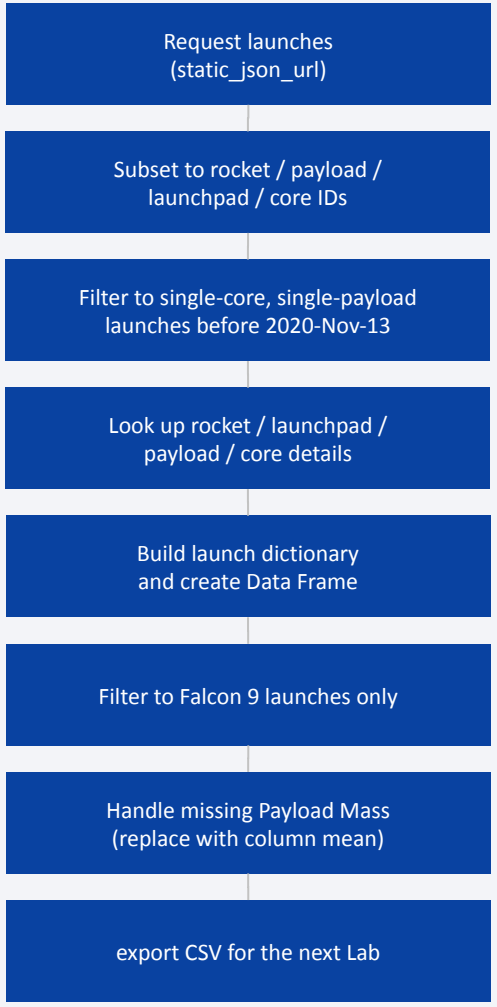
- **Data Collection:** Combined the SpaceX REST API and Wikipedia web scraping in one dataset to gather Falcon 9 launch records
- **Data Wrangling:** Filtered the dataset to Falcon 9 launches, handled missing payload values, and engineered the landing-success label
- **Exploratory Data Analysis (EDA) with Visualization & SQL:** Explored relationships between payload, orbit, launch site, and landing outcome
- **Interactive Visual Analytics:** Built a Folium map and Plotly Dash dashboard to explore the data visually
- **Predictive Analysis:** Trained and tuned four classification models to predict landing outcome

Data Collection

Launch data was collected from two complementary sources: the SpaceX REST API (api.spacexdata.com v4), and the Falcon 9/Falcon Heavy launch history table on Wikipedia (via web scraping)

Combining both sources in one data set let us cross-validate fields such as booster version, payload mass, orbit, and landing outcome across an independent, human-curated source

Data Collection – SpaceX API

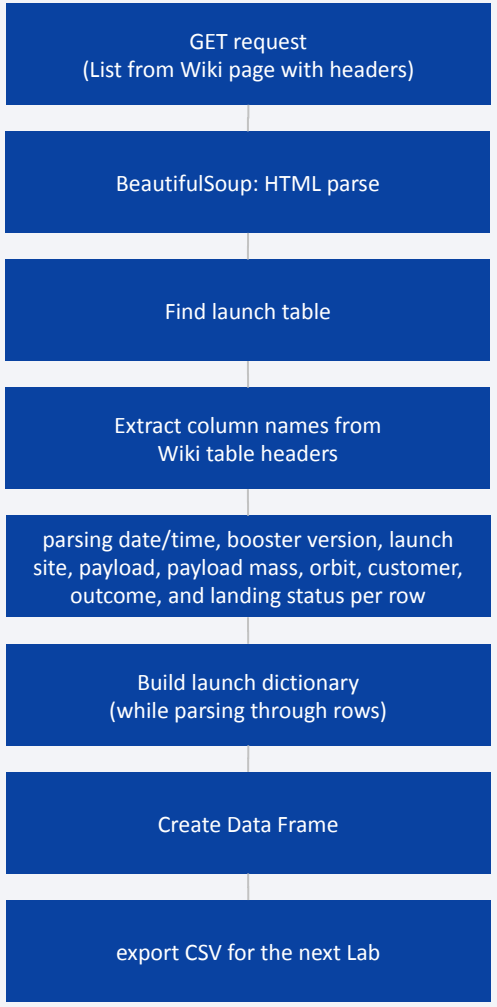


- Requested the launch list from the SpaceX v4 API (GET request to the /launches endpoint), then used the rocket, payload, launchpad, and core IDs in each record to request booster version, payload mass/orbit, launch site coordinates, and landing outcome details via further GET requests
- Subsetted rocket, payload, launchpad and core IDs, and filtered for given criteria to build the launch dictionary (launch_dict)
- Created a DataFrame based on launch_dict and filtered to Falcon 9 launch data only
- Missing PayloadMass values were replaced with the column mean

Note: the live per-launch endpoints (/v4/rockets/{id}, /v4/launchpads/{id}, /v4/payloads/{id}, /v4/cores/{id}) intermittently returned HTTP 525 errors (an SSL handshake failure on SpaceX's backend) during this project. Where this blocked completion, IBM's own static snapshot of the same, already-collected dataset (dataset_part_1.csv) was used instead. Usage of IBM's dataset is documented directly in the notebook

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/15d180ac554cdc80f0dbf2baaf4ea746acbaceae/jupyter-labs-spacex-data-collection-api(9)-Task1%20fixed.ipynb

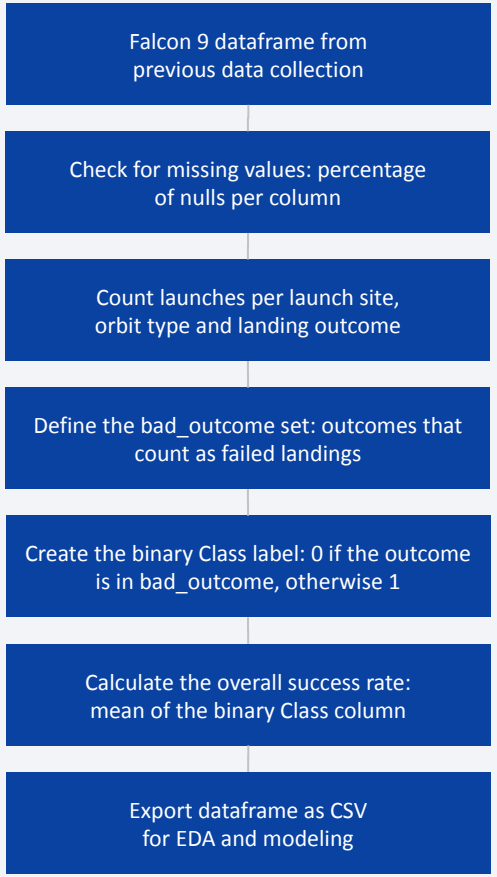
Data Collection - Web Scraping



- Requested the "List of Falcon 9 and Falcon Heavy launches" Wikipedia page with a browser User-Agent header
- Parsed the tables with BeautifulSoup
- Located the launch history table (3rd table on the Wiki page) and extracted the column names from the table headers
- Build the launch dictionary by parsing date/time, booster version, launch site, payload, payload mass, orbit, customer, outcome, and landing status
- Create the date frame and export it as CSV for the next section

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/15d180ac554cdc80f0dbf2baaf4ea746acbaceae/jupyter-labs-webscraping.ipynb

Data Wrangling



- The combined dataset of Falcon 9 launches from the previous data collection was used as starting point
- The dataset was checked for missing values and the launches were counted per launch site, orbit type and landing outcome
- Failed landing outcomes were defined as bad_outcome set
- Engineered the binary target label Class:
  - 1 for a successful first-stage landing outcome,
  - 0 for a landing outcome in the predefined bad_outcome set (e.g. failures and uncontrolled landings) – this
- The overall success rate was calculated based on the mean of the binary target label Class
- Binary Class column became the prediction target for the Machine Learning models in the following labs

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/d10634968fc176888825b5b1ec254b2a27637546/labs-jupyter-spacex-Data%20wrangling(1).ipynb

EDA with Data Visualization

- Used seaborn/matplotlib to create scatter plots, bar charts and line charts to explore how launch outcome relates to flight number, payload mass, launch site, and orbit
  - The scatter plots were used to reveal trends/clusters
    - was used to visualize flight number vs. payload mass, flight number vs. launch site, launch site vs. payload mass, flight number vs. orbit type and orbit vs. payload mass
  - A bar chart best compares a rate (e.g., success rate) across categories
    - was used to visualize success rate vs. orbit
  - A line chart best shows a trend over time
    - was used to visualize success rate depending on the year

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/15d180ac554cdc80f0dbf2baaf4ea746acbaceae/edadataviz(2).ipynb

EDA with SQL

- Loaded the wrangled dataset into a SQLite table (SPACEXTABLE)
- used SQL to directly answer specific questions:
    - unique launch sites
    - records from CCAFS pads
    - total NASA (CRS) payload mass
    - average F9 v1.1 payload mass
    - first successful ground-pad landing date
    - drone-ship successes within a payload range
    - mission outcome totals,
    - boosters with maximum payload
    - 2015 monthly failures
    - ranked count of landing outcomes over a date range

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/ae49cc14df56aee8cf1fac6ed0d5ce4cea5414ab/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

Added a Circle and text Marker for every launch site (color-coded, labeled with the site name) to show their global distribution; a MarkerCluster of individual launch markers colored green (success, Class=1) or red (failure, Class=0) at each site; and a Marker and PolyLine connecting a launch site to its nearest coastline point, with the great-circle distance calculated and displayed

These let us visually compare launch sites, see the mix of successes/failures at each one, and confirm that launch sites are built close to a coastline – useful for safety (launching over water) and for understanding site selection

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/ae49cc14df56aee8cf1fac6ed0d5ce4cea5414ab/lab_jupyter_launch_site_location(3).ipynb

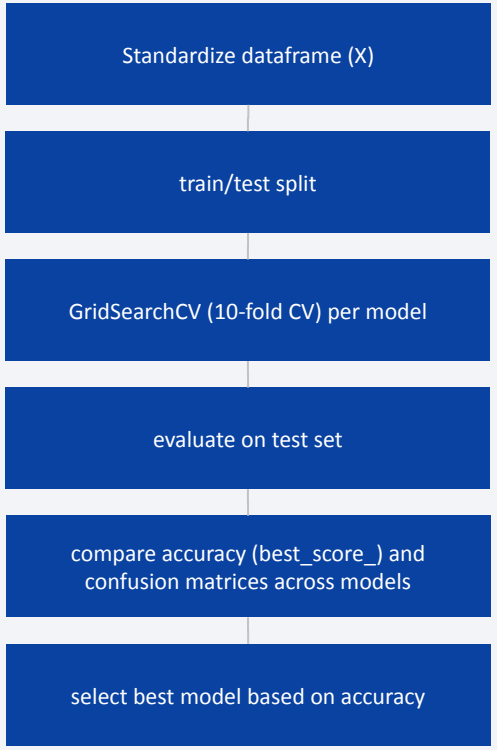
Build a Dashboard with Plotly Dash

Built a dropdown to select a launch site (or all sites), a pie chart showing total successful launches per site (or the success/failure split for one selected site, colored consistently red=failure/green=success), a payload-mass range slider, and a scatter plot of payload vs. launch outcome colored by booster version category

These let a user interactively compare launch-site success rates and see how payload mass and booster version relate to landing outcome, without needing to write any code

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/8bbce64e2a59ab7f53f8a114c53c22e09e71ff12/spacex-dash-app.py

Predictive Analysis (Classification)



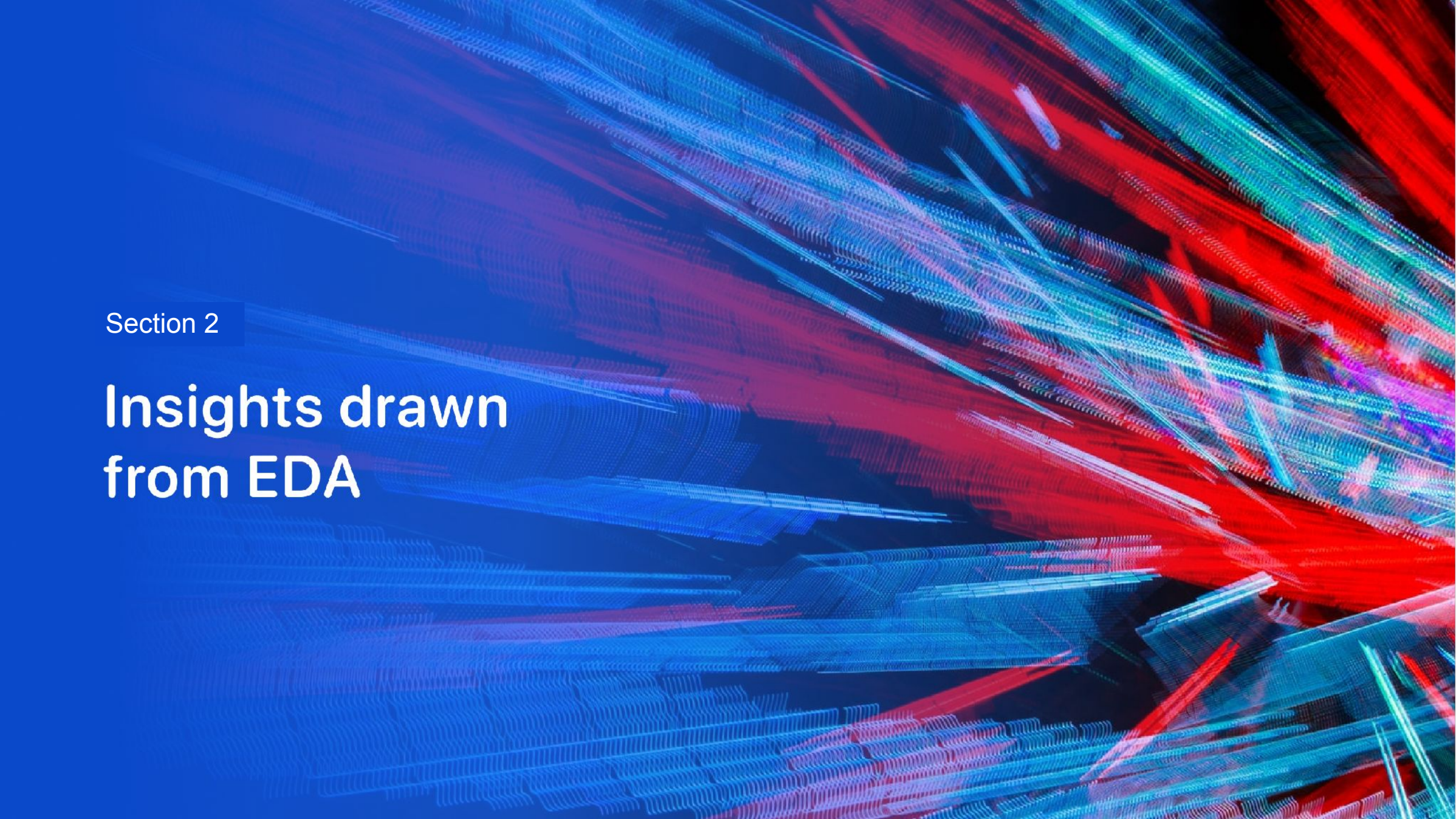
- Standardized the numeric/one-hot-encoded features with StandardScaler, then
- split the data 80/20 into training and test sets (random_state=2, 18 test records).
- Four classifiers (Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-nearest neighbor (KNN)) were each tuned via GridSearchCV with 10-fold cross-validation over a hyperparameter grid, and evaluated on the held-out test set

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/8a29d309b9dd9cc119ded48338c4b3e6103e8561/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

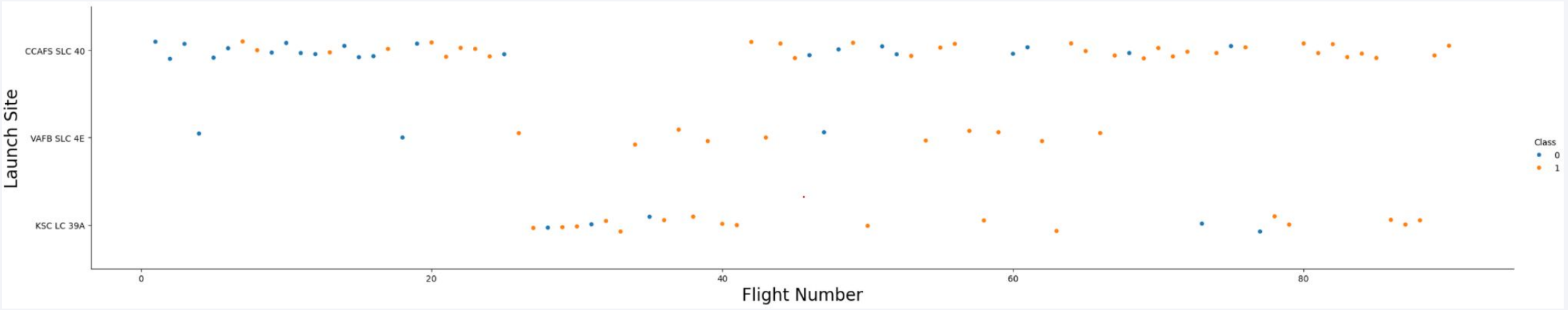
Results

Executive Summary

- **Exploratory data analysis results:** visualizations and SQL queries on launch sites, payload, orbits, and yearly trends
- **Interactive analytics demo in screenshots:** Folium map and Plotly Dash dashboard
- **Predictive analysis results:** comparison of four classification models, with the Decision Tree performing best


Section 2
Insights drawn from EDA

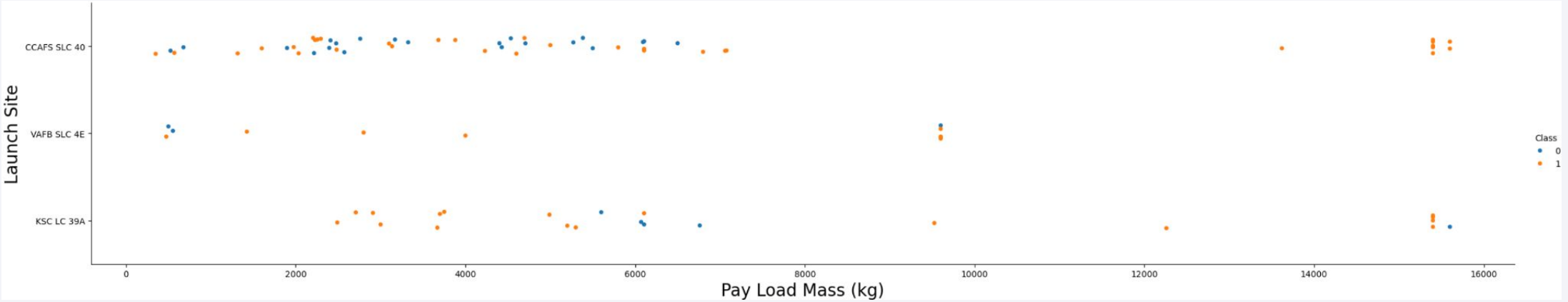
Flight Number vs. Launch Site



Earlier flight numbers show a higher frequency of unsuccessful landings (blue, Class 0), while later flights show a predominance of successful landings (orange, Class 1). This trend is particularly apparent at CCAFS SLC 40 and KSC LC 39A, suggesting an improvement in landing success as SpaceX's program matured. At VAFB SLC 4E, no equally clear trend is apparent, likely due in part to the smaller sample size of 13 launches.

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/ae49cc14df56aee8cf1fac6ed0d5ce4cea5414ab/edadataviz(2).ipynb

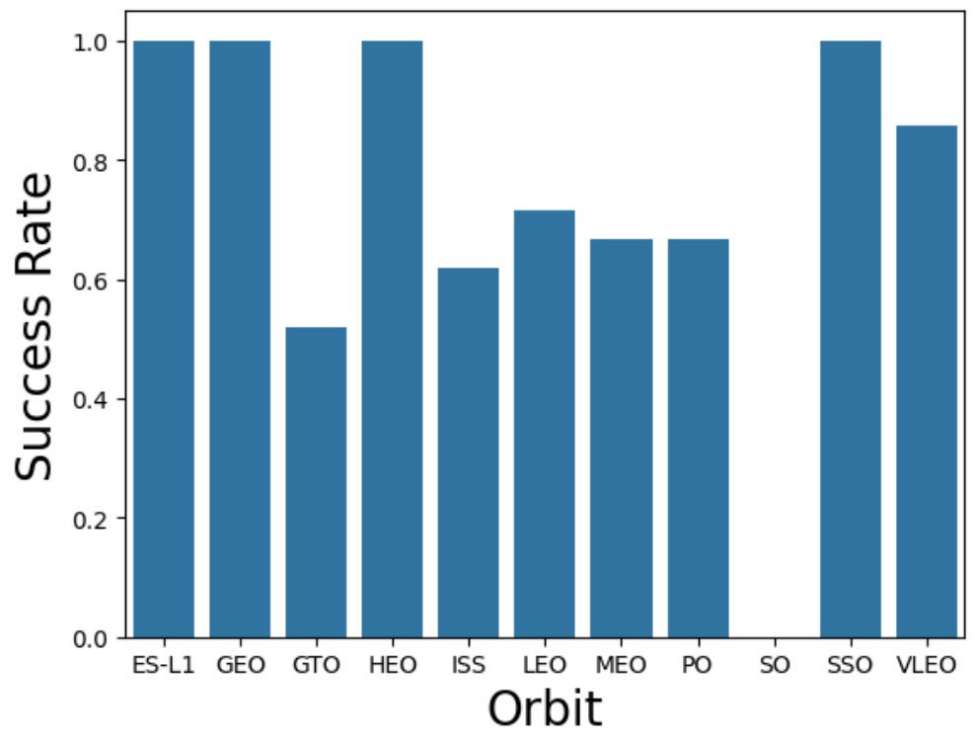
Payload vs. Launch Site



The launch site VAFB SLC-4E is not used for the heaviest payloads (above roughly 10,000 kg), while CCAFS and KSC LC-39A host launches across a wider payload range, including the heaviest. Lower payloads show mixed results of successful (orange, Class 1) and failed (blue, Class 0) landings, especially for CCAFS. The sample size for very heavy payloads is limited but payloads above roughly 10,000 kg appear to display fewer failures at the CCAFS and KSC LC-39A launch sites.

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/ae49cc14df56aee8cf1fac6ed0d5ce4cea5414ab/edadataviz(2).ipynb

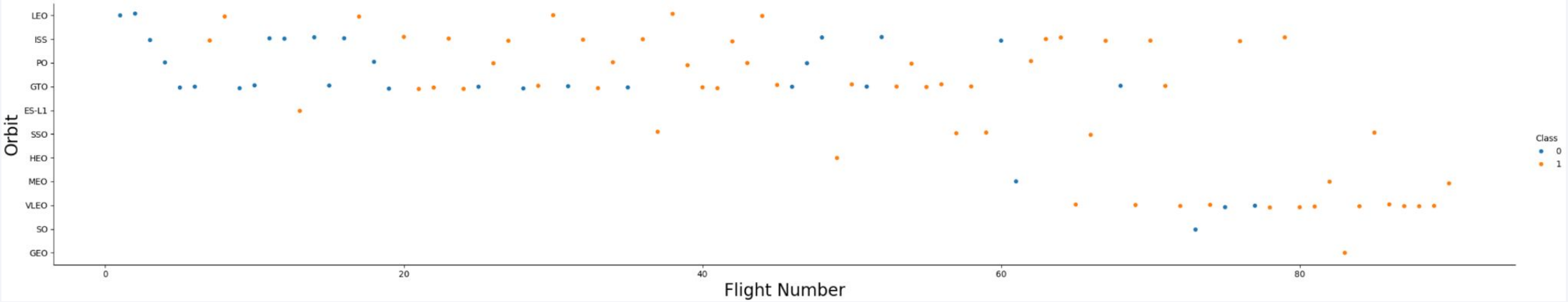
Success Rate vs. Orbit Type



At a first glance orbits such as ES-L1, GEO, HEO, and SSO show the highest success rates (at or near 100%), while GTO shows a comparatively lower success rate. Though ES-L1, GEO, HEO, and SSO orbits have very few launches. Five launches were performed for the SSO orbit and only one launch was performed to the ES-L1, GEO and HEO orbits, as well as the SO orbit which shows the lowest success rate (0%). Their rates should be read with that small sample size in mind and it could be considered to omit orbits with small sample sizes from the conclusions.

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/ae49cc14df56aee8cf1fac6ed0d5ce4cea5414ab/edadataviz(2).ipynb

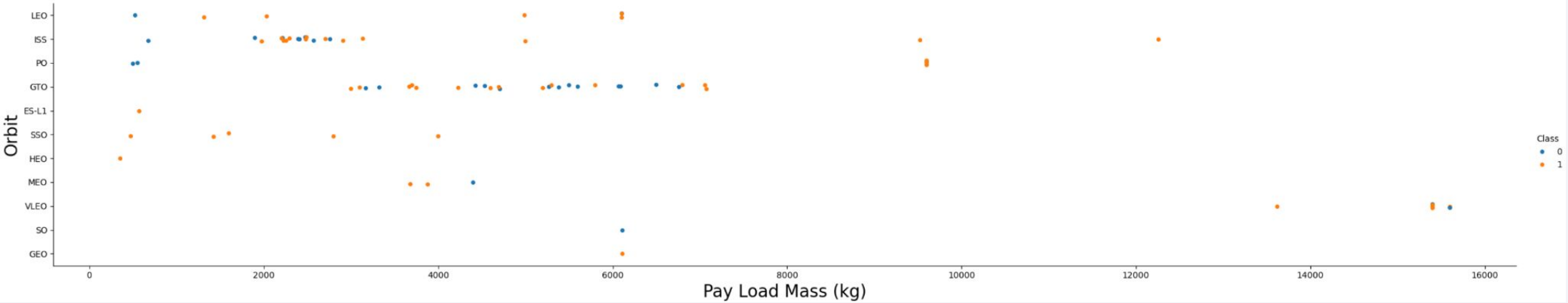
Flight Number vs. Orbit Type



The LEO, ISS, PO, and GTO orbits are represented across a broad range of flight numbers, while other orbit types, such as VLEO, appear mainly in later flights. Although successful landings (orange, Class 1) are more frequent among many higher-numbered flights, the relationship between flight number and landing success is not equally clear across orbit types. Differences in sample size should be considered when interpreting these patterns.

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/ae49cc14df56aee8cf1fac6ed0d5ce4cea5414ab/edadataviz(2).ipynb

Payload vs. Orbit Type

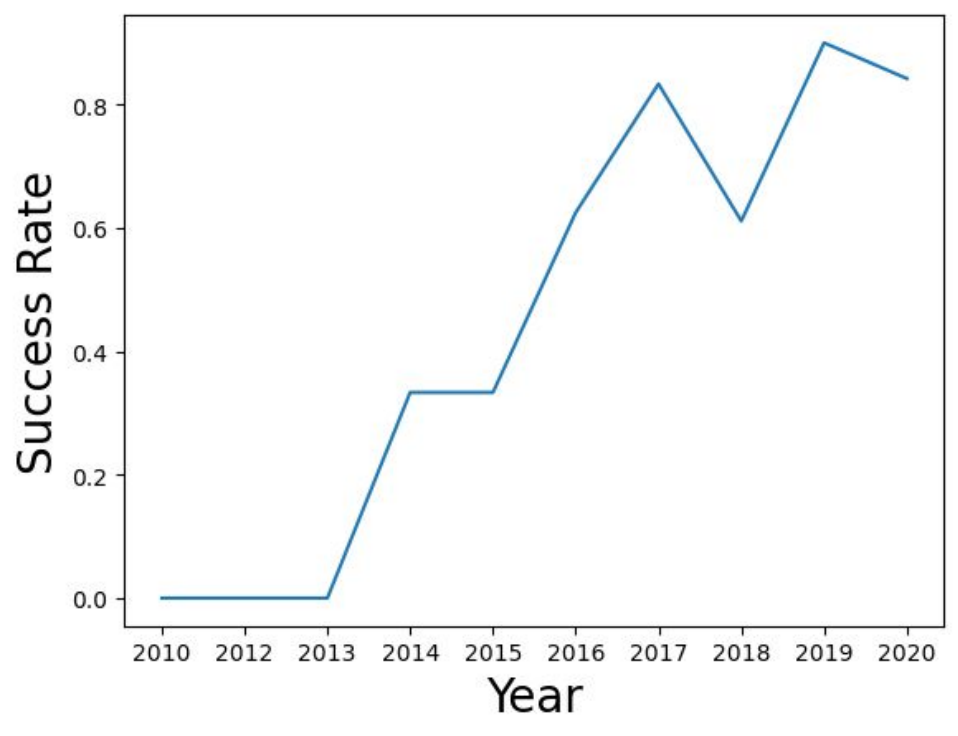


GTO missions span a wide moderate-to-high payload range, while LEO/ISS missions cluster at lower-to-moderate payload masses. Polar orbit (PO) payloads tend to be lighter, whereas VLEO payloads tend to be very heavy.

With heavy payloads the successful landing (orange, Class 1) or positive landing rate are more for PO, LEO and ISS although the limited sample size for heavier launches should be considered. For GTO, it's difficult to distinguish between successful and unsuccessful (blue, Class 0) landings as both outcomes are present over the payload range. A difficulty to distinguish between successful and unsuccessful landings can also be observed for lower-to-moderate payload masses of the ISS orbit.

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/ae49cc14df56aee8cf1fac6ed0d5ce4cea5414ab/edadataviz(2).ipynb

Launch Success Yearly Trend



The first launches in 2010 to 2013 were unsuccessful, which is reflected by a success rate of 0. The average success rate shows a clear upward trend from 2013 through 2020, with a dip in 2018. The early years from 2013 to 2016 are showing low/inconsistent success (<0.5) and later years approaching a higher success rate of >0.5 up to approximately 0.95 in 2019.

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/ae49cc14df56aee8cf1fac6ed0d5ce4cea5414ab/edadataviz(2).ipynb

All Launch Site Names

- **Task:** Find the names of the unique launch sites
- **Query:** %sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;
- **Result:** returns the four launch sites used across all Falcon 9 missions in the dataset

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/ae49cc14df56aee8cf1fac6ed0d5ce4cea5414ab/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Launch Site Names Begin with 'CCA'

- **Task:** Find 5 records where launch sites begin with `CCA`
- **Query:** %sql SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
- **Result:** selects 5 launch sites starting with CCA

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/ae49cc14df56aee8cf1fac6ed0d5ce4cea5414ab/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Total Payload Mass

- **Task:** Calculate the total payload carried by boosters from NASA
- **Query:** %sql SELECT SUM("PAYLOAD_MASS__KG_") FROM SPACEXTABLE WHERE "Customer" = 'NASA (CRS)';
- **Result:** sum of the payload carried by boosters from NASA in kg

SUM("PAYLOAD_MASS__KG_")
45596

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/ae49cc14df56aee8cf1fac6ed0d5ce4cea5414ab/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Average Payload Mass by F9 v1.1

- **Task:** Calculate the average payload mass carried by booster version F9 v1.1
- **Query:** %sql SELECT AVG("PAYLOAD_MASS__KG_") FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1';
- **Result:** the average payload mass across all launches using the F9 v1.1 booster version in kg

AVG("PAYLOAD_MASS__KG_")
2928.4

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/ae49cc14df56aee8cf1fac6ed0d5ce4cea5414ab/jupyter-labs-eda-sql-coursera_sqllite.ipynb

First Successful Ground Landing Date

- **Task:** Find the date of the first successful landing outcome on a ground pad
- **Query:** %sql SELECT MIN("Date") FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';
- **Result:** the date of the first-ever successful Falcon 9 ground-pad landing, a key milestone in booster reusability

MIN("Date")
2015-12-22

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/ae49cc14df56aee8cf1fac6ed0d5ce4cea5414ab/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Successful Drone Ship Landing with Payload between 4000 and 6000

- **Task:** List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000
- **Query:** %sql SELECT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)' AND "PAYLOAD_MASS__KG_" > 4000 AND "PAYLOAD_MASS__KG_" < 6000;
- **Result:** booster versions that successfully landed on a drone ship while carrying a mid-range payload (4,000-6,000 kg)

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/ae49cc14df56aee8cf1fac6ed0d5ce4cea5414ab/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Total Number of Successful and Failure Mission Outcomes

- **Task:** Calculate the total number of successful and failure mission outcomes
- **Query:** %sql SELECT "Mission_Outcome", COUNT(*) FROM SPACEXTABLE GROUP BY "Mission_Outcome";
- **Result:** the total count of each mission outcome category, showing the overall mission-level (not landing-level) success rate

Mission_Outcome	COUNT(*)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/ae49cc14df56aee8cf1fac6ed0d5ce4cea5414ab/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Boosters Carried Maximum Payload

- **Task:** List the names of the booster which have carried the maximum payload mass
- **Query:** %sql SELECT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS__KG_" = (SELECT MAX("PAYLOAD_MASS__KG_") FROM SPACEXTABLE);
- **Result:** the booster versions carrying the maximum payload mass in the dataset

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/ae49cc14df56aee8cf1fac6ed0d5ce4cea5414ab/jupyter-labs-eda-sql-coursera_sqllite.ipynb

2015 Launch Records

- **Task:** List the failed landing_outcomes on drone ships, their booster versions, and launch site names for in year 2015
- **Query:** %sql SELECT substr("Date",6,2) as month, "Landing_Outcome", "Booster_Version", "Launch_Site" FROM SPACEXTABLE WHERE substr("Date",0,5)='2015' AND "Landing_Outcome" = 'Failure (drone ship)';
- **Result:** the month, booster version, and launch site for every 2015 launch that failed to land on a drone ship, useful for spotting any seasonal or booster-specific pattern in early landing attempts

month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/ae49cc14df56aee8cf1fac6ed0d5ce4cea5414ab/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

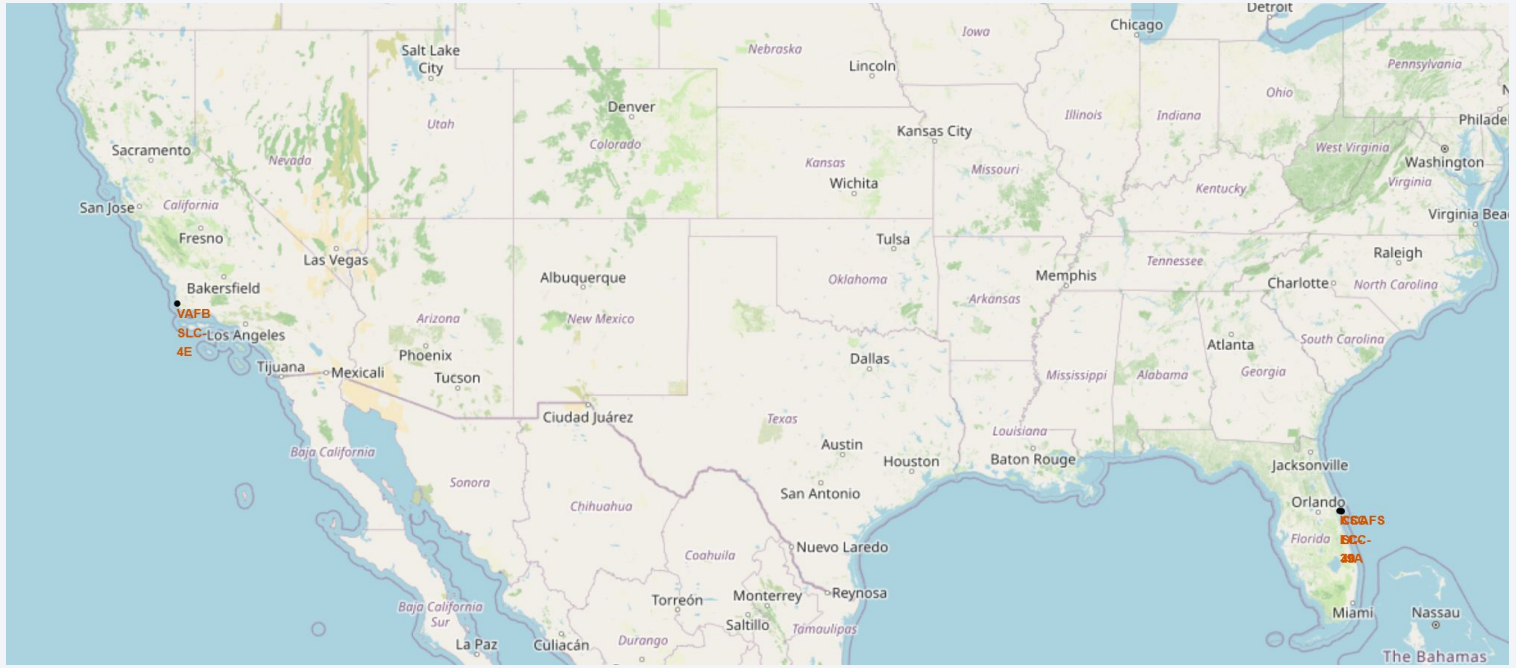
- **Task:** Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order
- **Query:** %sql SELECT "Landing_Outcome", COUNT(*) as outcome_count FROM SPACEXTABLE WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY "Landing_Outcome" ORDER BY outcome_count DESC;
- **Result:** landing outcomes ranked from most to least common during this early period of the program, showing failures still outnumbering successes before landing technology matured

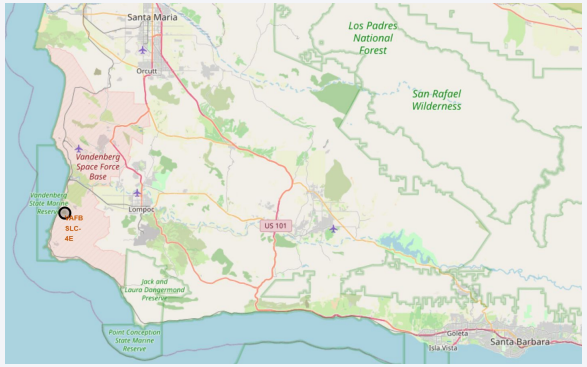
Landing_Outcome	outcome_count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

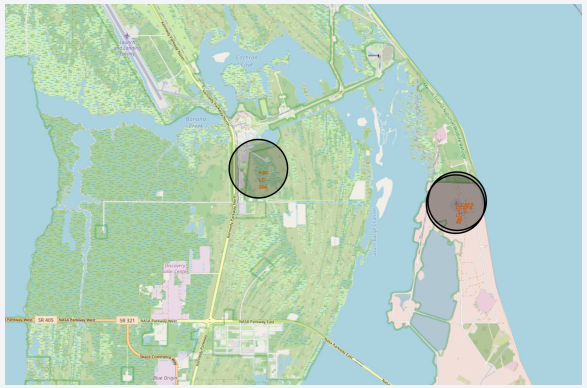
GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/ae49cc14df56aee8cf1fac6ed0d5ce4cea5414ab/jupyter-labs-eda-sql-coursera_sqllite.ipynb


Section 3
Launch Sites
Proximities Analysis

All Launch Sites on a Global Map



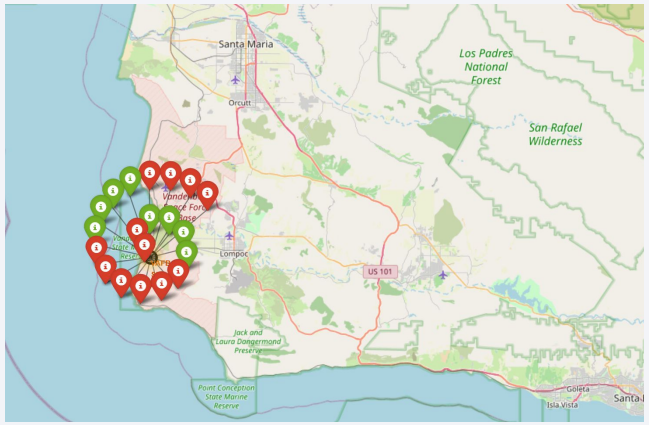


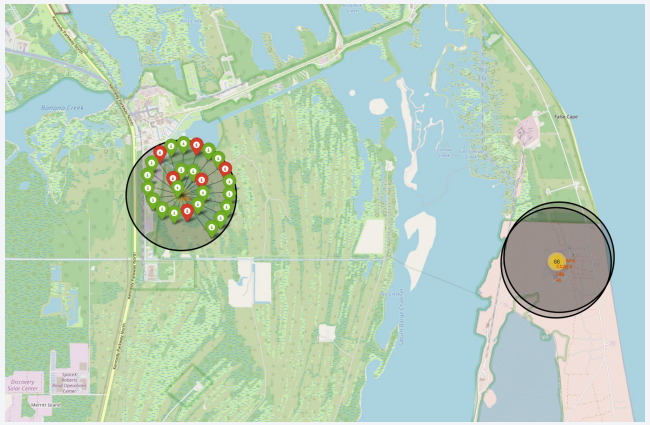


All four Falcon 9 launch sites - CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, and VAFB SLC-4E - are shown as circle markers with popup labels. All three East Coast sites cluster tightly together in Florida, while VAFB SLC-4E sits alone on the California coast, confirming SpaceX's two main launch corridors.

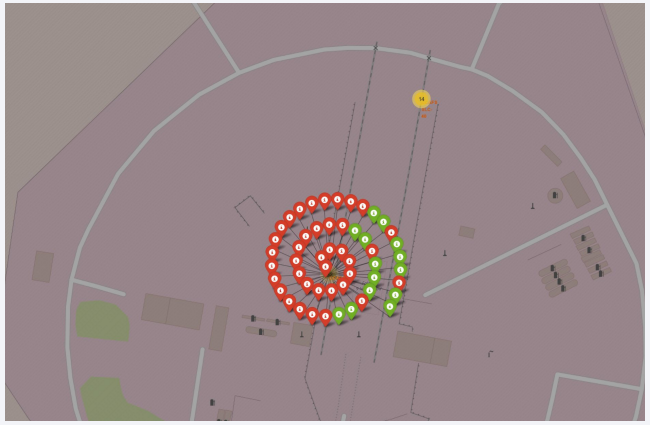
GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/ae49cc14df56aee8cf1fac6ed0d5ce4cea5414ab/lab_jupyter_launch_site_location(3).ipynb

Launch Outcomes by Site





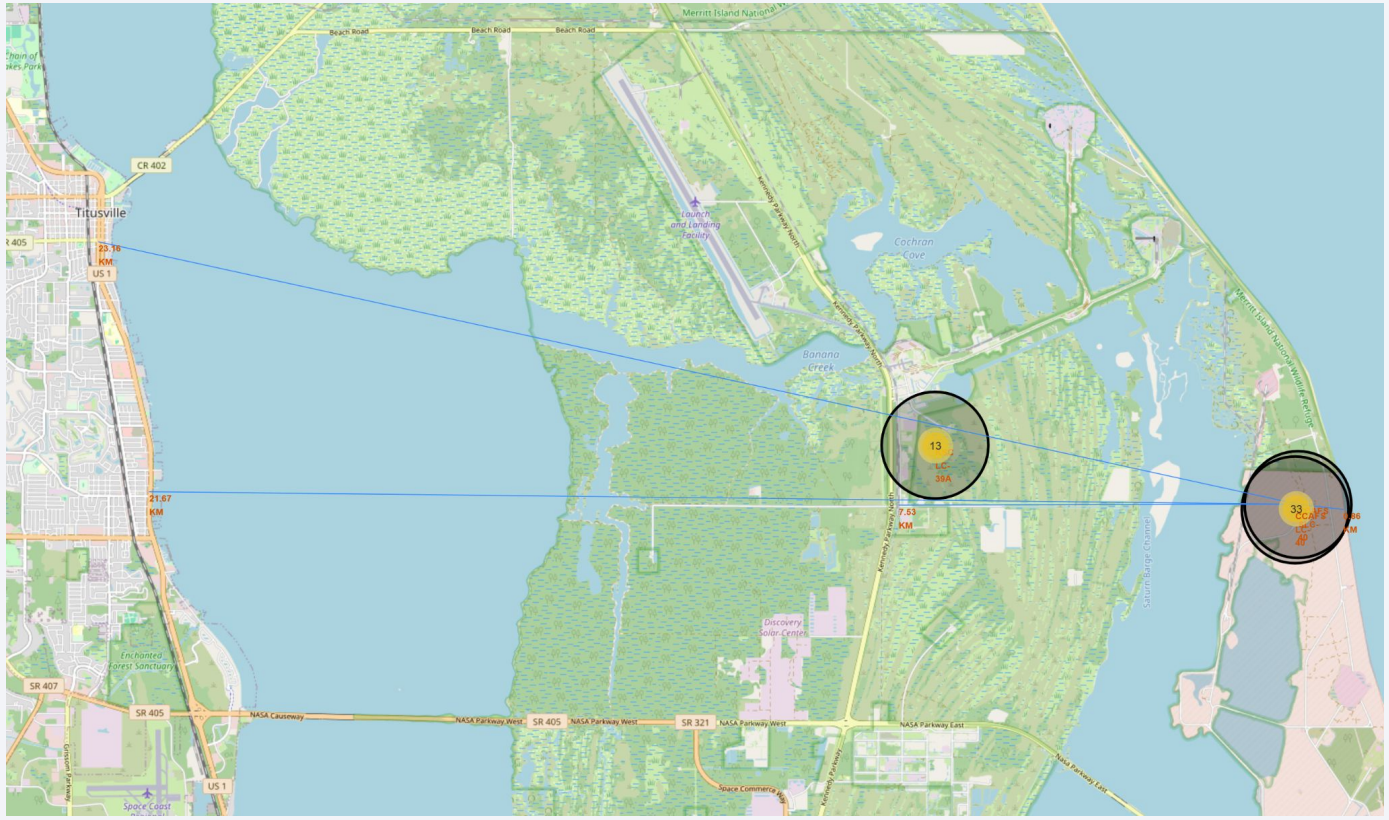




Each individual launch is plotted as a marker within its site's cluster, colored green for a successful landing (Class=1) and red for a failed landing (Class=0). This makes it easy to compare the mix of outcomes across sites at a glance.

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/ae49cc14df56aee8cf1fac6ed0d5ce4cea5414ab/lab_jupyter_launch_site_location(3).ipynb

Launch Site Proximity to nearby Features



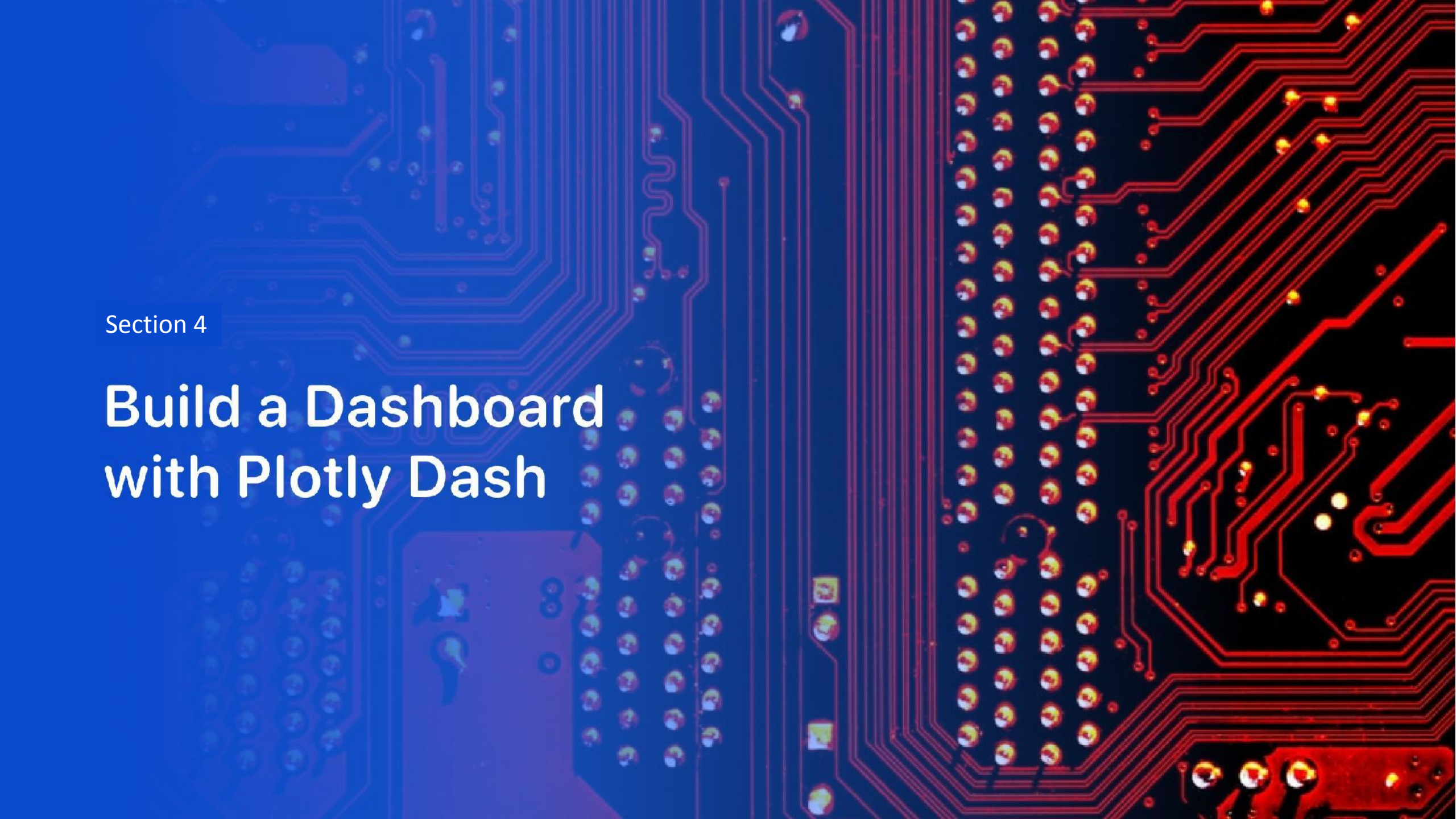
A marker and connecting line were added between a launch site and its nearest coastline point, railway, highway and city. The calculated great-circle distance was displayed on the marker.

In general launch sites are built close to the coast so that the rocket's flight path, and any failure, occurs over water rather than populated land. Railways and smaller streets are close to the launch sites to transport equipment, rocket stages, and staff to the site. Generally cities and US highways are further away to keep people safe in case of an accident. For example the closest US highway to the SpaceX launch sites in Cape Canaveral is US1, which runs through the nearest city Titusville.

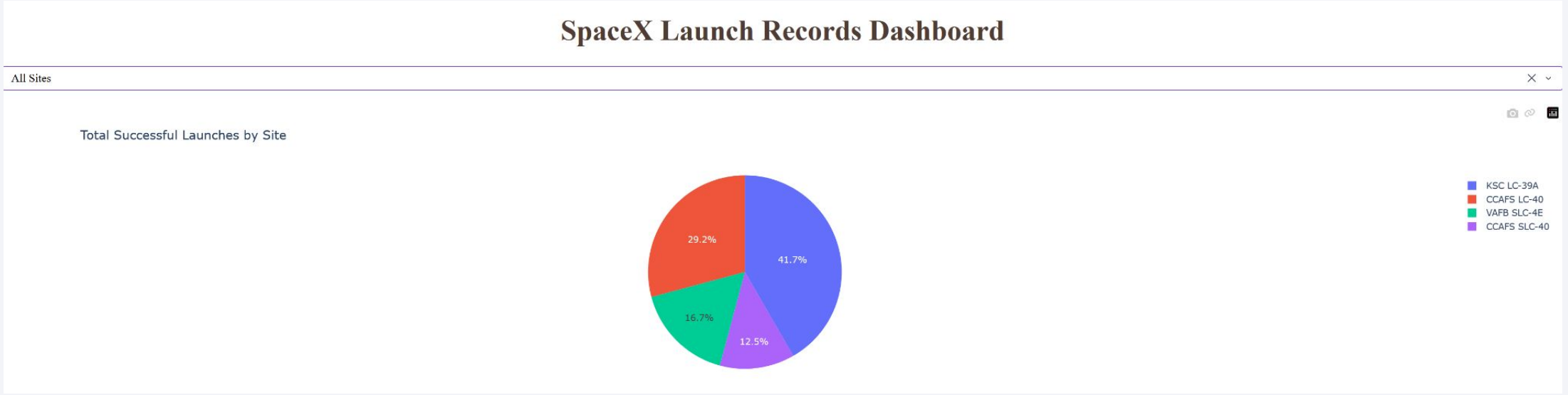
GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/ae49cc14df56aee8cf1fac6ed0d5ce4cea5414ab/lab_jupyter_launch_site_location(3).ipynb

Section 4

Build a Dashboard with Plotly Dash



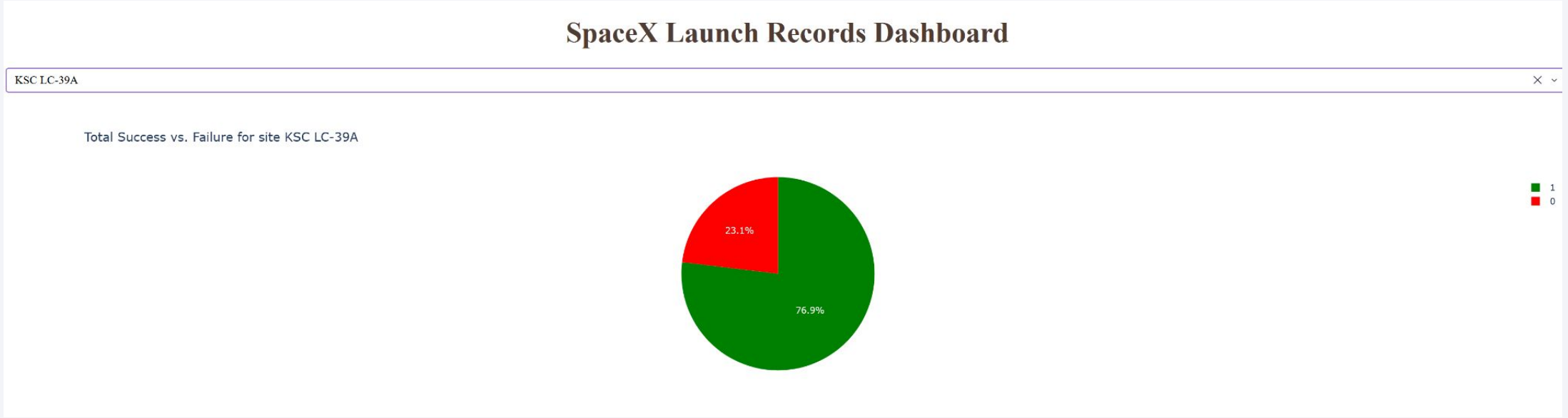
Launch Success of the Sites



With “All Sites” selected, the pie chart shows each launch site's share of all successful launches recorded in the dataset. KSC LC-39A accounts for the largest share (41.7%), followed by CCAFS LC-40 (29.2%), VAFB SLC-4E (16.7%) and CCAFS SLC-40 (12.5%). These percentages represent contributions to the total number of successful launches, rather than the success rate at each site.

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/ae49cc14df56aee8cf1fac6ed0d5ce4cea5414ab/spacex-dash-app.py

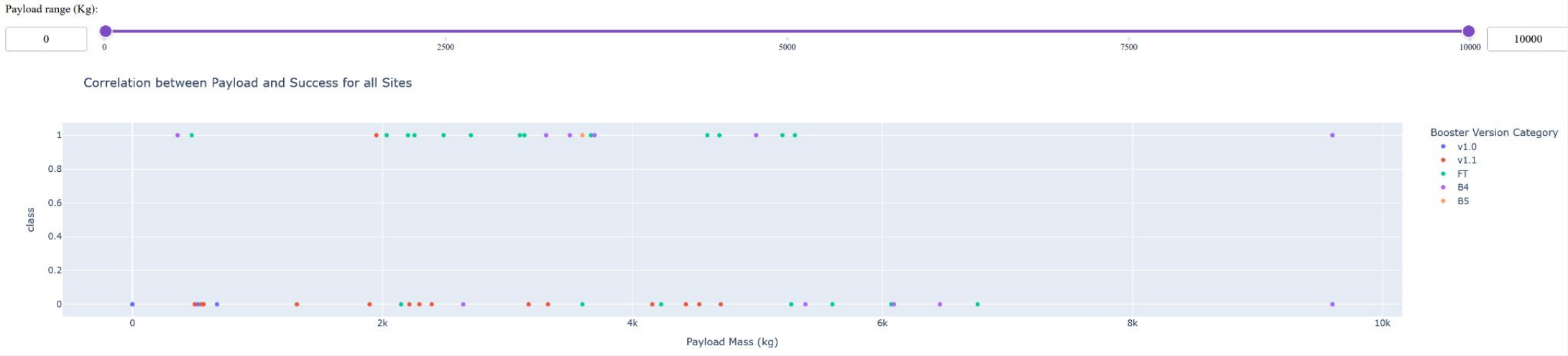
Success vs. Failure ratio for the Top-Performing Site KSC LC-39A



Filtering the dashboard by launch site reveals the proportion of successful and unsuccessful landing outcomes at each site. KSC LC-39A has the highest observed landing success rate, at 76.9%, compared with a failure rate of 23.1%. The other listed sites have lower observed success rates: CCAFS SLC-40 (42.9%), VAFB SLC-4E (40.0%) and CCAFS LC-40 (26.9%). These results describe the recorded outcomes in the dataset and do not necessarily account for differences in mission characteristics or sample size.

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/ae49cc14df56aee8cf1fac6ed0d5ce4cea5414ab/spacex-dash-app.py

Payload Mass vs. Launch Outcome by Booster Version



The scatter plot colors each launch point by booster version category, plotted against payload mass and outcome (Class). Adjusting the payload range slider shows that lower-to-moderate payload masses (2000 to 6000 kg) are associated with a higher success rate, and that certain booster version categories (FT and B4) show consistently higher success across the payload range than older versions (v1.0 and v1.1).

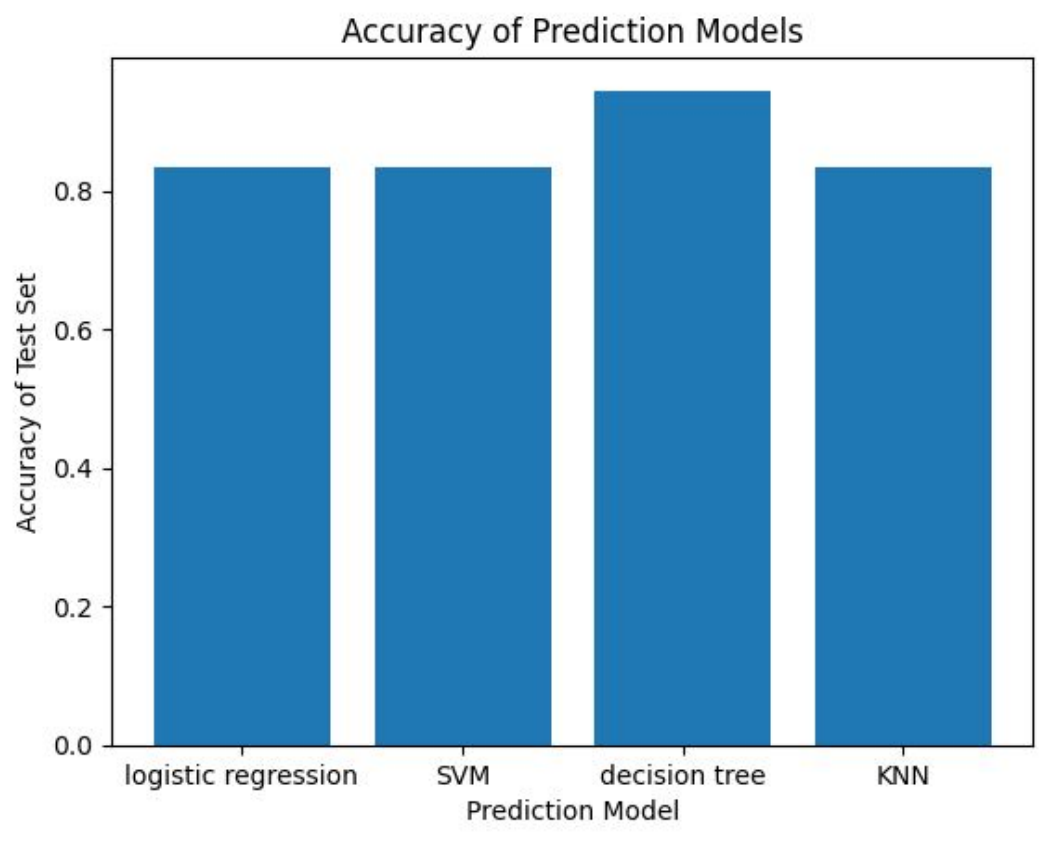
GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/ae49cc14df56aee8cf1fac6ed0d5ce4cea5414ab/spacex-dash-app.py

Section 5

Predictive Analysis (Classification)



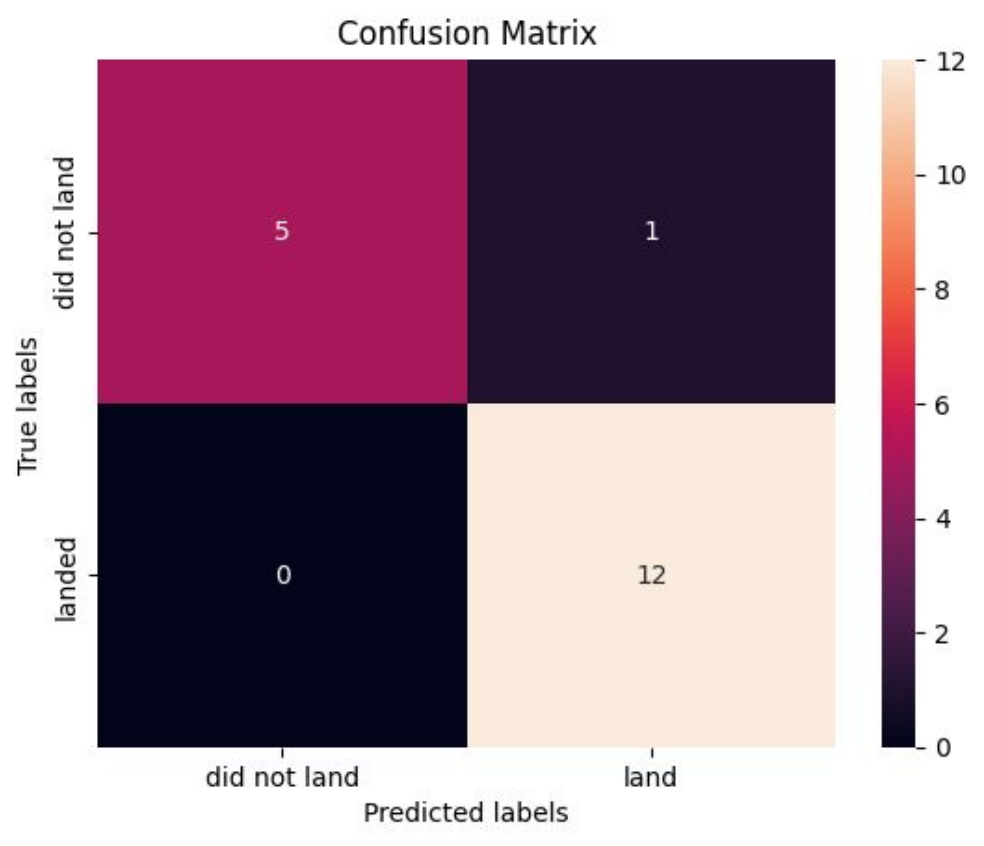
Classification Accuracy



- The bar chart compares the test-set accuracy of the four classification models.
- While some models show higher accuracy than others, the differences should be interpreted with caution because the test set contains only 18 observations.
- Each individual prediction therefore has a noticeable impact on the accuracy score.
- The results provide an initial comparison of model performance, but further validation on a larger dataset would be needed to determine whether the observed differences are meaningful.

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/8a29d309b9dd9cc119ded48338c4b3e6103e8561/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Confusion Matrix-Decision Tree



- The Decision Tree model correctly classified 17 of the 18 test samples, achieving an accuracy of 94.4%. It correctly identified 5 unsuccessful landings and 12 successful landings, with only one incorrect prediction.
- The model produced no false negatives, meaning it did not misclassify any successful landings as unsuccessful in this test set. However, it incorrectly predicted one unsuccessful landing as successful. Given the small test set, these results should be interpreted with caution.

GitHub URL:
https://github.com/cloudychan143/IBM-Applied-Data-Science-Capstone-/blob/8a29d309b9dd9cc119ded48338c4b3e6103e8561/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Conclusions

- Falcon 9 first-stage landing outcomes can be predicted using pre-launch mission parameters, including payload mass, orbit, launch site, and booster version. The Decision Tree's achieved 86.25% cross-validated accuracy, indicating promising predictive performance using information available before launch.
- The Decision Tree achieved the highest observed performance (94.4%) among Logistic Regression, SVM, and KNN on this dataset. However, differences should be interpreted cautiously because the test set contained only 18 records.
- Landing outcomes show an overall improvement across the Falcon 9 program, with successful landings becoming more frequent in later flights. This pattern is consistent with progress in booster recovery, although the analysis alone cannot establish the causes of the improvement.
- Landing outcomes varied with payload mass and orbit type. Some orbit categories, including some less-represented orbit categories showed high observed success, while GTO missions displayed greater variability. These patterns should be interpreted in light of differences in sample size and mission characteristics.

Thank you!

